



INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Supply Chain Visibility & Optimization

- Turning raw logistics data into real-time, data-driven decision

MileStone-2

Presented by:

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Title: Supply Chain Visibility s Optimization

1.Objective

The main objective of the Supply Chain Visibility & Optimization project is to analyze supply-chain data and improve operational performance through data-driven insights. The project focuses understanding inventory, products, sales, revenue, suppliers, transportation, fulfillment, shipping and quality information. Python is used for data preprocessing and exploratory analysis, while Power BI and DAX are used to create KPIs and interactive dashboards. The analysis helps identify inventory and logistics patterns, evaluate product and supplier performance, monitor fulfillment and quality issues, and provide useful recommendations for optimizing supply-chain operations.

2.Introduction

The Supply Chain Visibility & Optimization project is a data analytics project focused on understanding and improving supply-chain operations. It analyzes information related to products, inventory, sales, suppliers, transportation, shipping, fulfillment and quality. Python is used for data preprocessing and exploratory data analysis, while Power BI and DAX are used to create KPIs, interactive dashboards and visualizations. The project helps identify important patterns in inventory, product performance, supplier contribution, logistics and quality, enabling better decision-making and supporting the optimization of overall supply-chain performance.

3. Milestone 2 Goals

- Transform processed data into understandable business information.
- Study inventory, revenue, product movement and order activity.
- Measure operational performance through suitable KPIs.
- Build Power BI reports that allow users to explore the data interactively.
- Identify supplier, transportation and quality-related patterns.

- Highlight operational areas that may need management attention.
- Support decisions using visual evidence rather than raw records.

4. Data Used for Analysis

The project dataset contains 1,540 records and 24 attributes. It combines information from different parts of the supply chain, making it possible to examine inventory, products, sales, suppliers, manufacturing, shipping and inspection activities.

- SKU and product category information
- Price, availability and stock-level information
- Products sold, order quantity and revenue information
- Supplier and manufacturing information
- Shipping carrier, route and transportation information
- Shipping and delivery duration
- Inspection outcomes and defect rates

5. Analysis Approach

The analysis was organized into a sequence that converts raw information into useful decisions:

Processed Dataset → Data Exploration → KPI Identification → Visual Analysis → Pattern Identification → Business Recommendations

The main areas examined were inventory utilization, product movement, supplier contribution, revenue, fulfillment, transportation, shipping performance and product quality.

- Distribution of inventory
- Comparison of product categories

- Supplier-level contribution
- Order and fulfillment behavior
- Transportation-mode distribution

6. Performance Measures

The following measures were selected because they provide a compact view of supply-chain performance.

Measure	What it helps to understand
Total Inventory	How much stock is currently available.
Inventory Value	The monetary value represented by the available stock.
Inventory Turnover	How efficiently stock is being utilized.
Total Revenue	Overall revenue generated from the analyzed products.
Total Orders	Volume of orders handled.
Fulfillment Rate	The level of successful order fulfillment.
Average Shipping Time	Typical time required to ship products.
Average Delivery Time	Typical time taken for delivery.

7. Dashboard A – Supply Chain Performance

The first Power BI page is designed as a management overview. KPI cards provide the important figures at a glance, while supporting charts explain how inventory, revenue, fulfillment and logistics are distributed.

1.Main Results

The main results of the project show that the supply chain has 75.62K total inventory with an inventory value of 3.75M and an inventory turnover of 9.39. The analysis recorded 8.84M total revenue and around 2K total orders, with a 90.38% fulfillment rate. The average shipping and delivery time is 5.74 days. Fast-moving and slow-moving products are almost equally distributed, with 768 fast-moving and 767 slow-moving products. Supplier 1 has the highest inventory-value contribution, while Cosmetics show comparatively higher defect rates. Also, approximately 45.15% of inspections are pending, indicating a need for faster inspection processing. Road transportation has the largest inventory share at approximately 41.37%.

Indicator	Observed Value
Available Inventory	75.62K
Inventory Value	3.75M
Inventory Turnover	9.39
Fast-Moving Products	768
Slow-Moving Products	767
Revenue	8.84M
Average Shipping Time	5.74 days
Average Delivery Time	5.74 days
Orders	2K
Fulfillment	90.38%

2.Visual Analysis Included

- Inventory distribution across product categories
- Revenue contribution by product category
- Fulfillment comparison across transportation modes
- Location-wise stock distribution

- SKU-level stock comparison
- Shipping and delivery time comparison
- Interactive slicers for supplier, location, product type, route, carrier, inspection and transportation mode

8. Dashboard B – Product, Supplier and Quality Review

The second Power BI page moves from the overall view to more detailed comparisons. It is intended to help identify product movement, supplier dependence, inspection workload and quality concerns.

1.Product Movement by Location

Fast-moving products are compared across locations to understand where product demand is relatively stronger. This can support more informed stock allocation and replenishment decisions.

2.Supplier Contribution

Inventory value is examined supplier by supplier. Supplier 1 contributes the largest inventory value, which makes supplier monitoring important when considering dependency and continuity of supply.

3.Inspection Status

Orders are categorized according to inspection outcomes. Pending inspections represent approximately 45.15% of the total, showing that inspection completion is an important area for operational improvement.

4.Quality by Product Category

Defect-rate values are compared for Skincare, Haircare and Cosmetics. Cosmetics show comparatively higher defect levels, making quality investigation and process improvement a priority.

5.Transportation Distribution

Inventory is compared across Road, Rail, Air and Sea. Road transportation

accounts for approximately 41.37% of the inventory share, making its performance important to overall logistics monitoring.

6.Supplier Sales Comparison

The number of products sold is compared across suppliers. Combining sales information with inventory contribution gives a broader view of supplier performance.

7.SKU-Level Information

- SKU
- Product type
- Price
- Number of products sold
- Revenue generated
- Stock levels

9. Major Observations

1.Stock Position

The analysis records approximately 75.62K units of inventory with an inventory value of about 3.75M. The turnover value of 9.39 indicates active inventory utilization. Fast-moving and slow-moving products are almost equally represented.

2.Product Category Observation

Skincare has a strong contribution to inventory and revenue. Haircare demonstrates comparatively steady sales, while Cosmetics need greater quality attention because of their higher defect-rate values.

3.Supplier Observation

Supplier 1 has the highest inventory-value contribution. Depending heavily a

major supplier can increase operational risk, so supplier performance should be reviewed continuously.

4.Logistics Observation

Shipping and delivery both average 5.74 days. Fulfillment is 90.38%, which indicates good overall order completion. Road transportation has the largest inventory share at 41.37%.

5.Inspection and Quality Observation

The pending inspection share is approximately 45.15%. Faster inspection handling could improve visibility of quality status. Cosmetics also require additional quality monitoring.

10. Suggested Improvements

- Set regular stock-review practices for high-value and high-volume products.
- Use product-movement categories to support replenishment and distribution planning.
- Study the causes of defects in Cosmetics and strengthen quality-control checks.
- Reduce the number of pending inspections by improving inspection turnaround.
- Track high-contribution suppliers and maintain alternatives where appropriate.
- Compare transportation performance to identify more efficient logistics choices.
- Monitor fulfillment and delivery measures as part of routine operational reviews.
- Add threshold-based KPI alerts in future versions of the dashboard.

11. Value of the Dashboard

The dashboard gives users a consolidated view of supply-chain performance instead of requiring them to examine individual records manually. It supports inventory monitoring, product comparison, supplier evaluation, quality review, fulfillment tracking and logistics analysis.

Interactive filtering also allows users to examine the same KPI from different perspectives such as product type, location, supplier, route or transportation mode.

11. Possible Extensions

- Predict future product demand using historical patterns.
- Develop models for identifying supplier risk.
- Create automated or near-real-time reporting.
- Predict possible stock shortages.
- Identify products that may become excess inventory.
- Generate automatic alerts when KPIs cross defined limits.
- Use analytics to improve transportation planning.

12. Conclusion

The Supply Chain Visibility & Optimization project successfully transformed supply-chain data into meaningful insights using Python, DAX and Power BI. The analysis helped evaluate inventory, revenue, product performance, supplier contribution, fulfillment, transportation and quality. The results show 75.62K total inventory, 3.75M inventory value, 8.84M revenue and a 90.38% fulfillment rate, with an average shipping time of 5.74 days. The dashboards also identified areas that need attention, such as pending inspections, higher defect rates in Cosmetics, supplier dependency and transportation optimization.